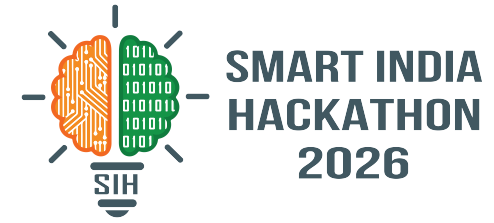
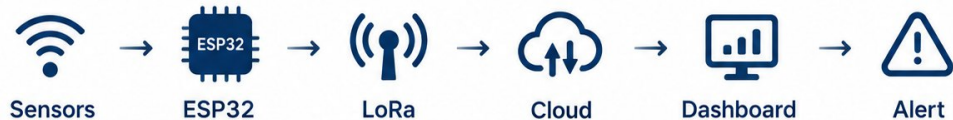
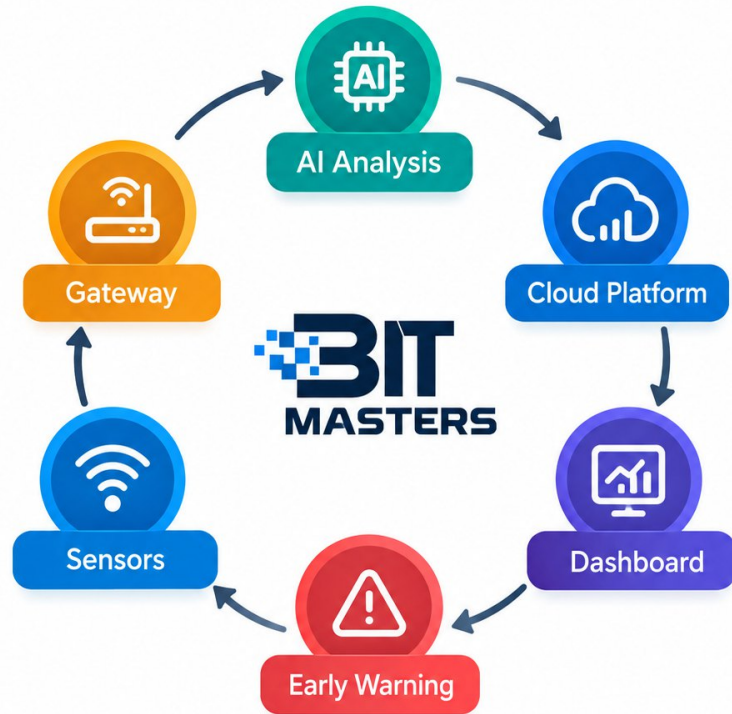


SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26025**
- **Problem Statement Title -AI-Enabled Real-Time Mine Subsidence Monitoring & Early Warning System**
- **Theme – Disaster Management**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name : Bit Masters**





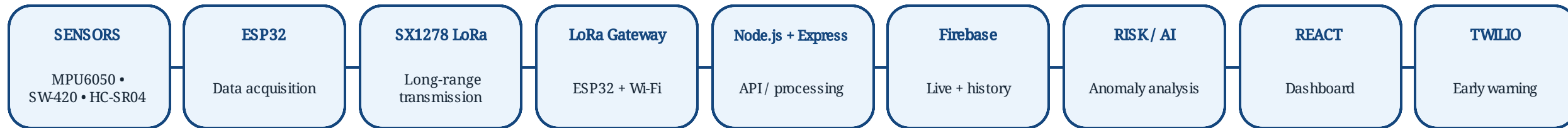
Current Problems

- Difficult to monitor mine subsidence **continuously in real time**
- Early ground deformation may go **undetected**
- Existing monitoring can be **costly and periodic**
- Delayed detection can increase **safety risks**

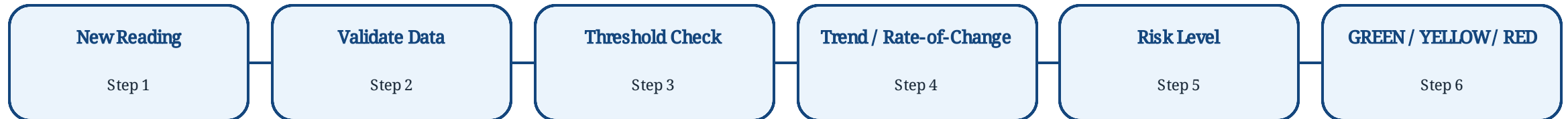
Our Idea

- **Low-cost IoT sensor nodes** for continuous monitoring
- **LoRa communication** for long-range, low-power data transmission
- Monitor **tilt, vibration and ground movement**
- **AI/ML-based analysis** to identify abnormal patterns
- **Real-time dashboard** for monitoring and visualization
- **Early warning alerts** for potential subsidence

SYSTEM DATA FLOW



RISK ANALYSIS LOGIC



FEASIBILITY & VIABILITY

- Low-cost ESP32 + LoRa hardware
- Open-source software & ML tools
- Low-power, scalable deployment

TECHNICAL CHALLENGES

- Reliable underground communication
- Sensor noise & missing data
- Accurate anomaly / subsidence prediction

USER & OPERATIONAL CHALLENGES

- Worker safety with minimal disruption
- Training & system awareness
- False alarms and maintenance

STRATEGIC APPROACH

- Pilot deployment in a limited area
- Modular architecture for expansion
- Improve ML using real-world data

1 EXISTING SOLUTIONS vs OUR APPROACH

Conventional Surveying



- Periodic measurements
- Limited real-time visibility

Advanced Monitoring



- High accuracy
- Higher infrastructure and deployment cost

Our Approach



- Low-cost distributed sensing
- LoRa-based communication
- Real-time dashboard
- Automated early-warning alerts

2 TECHNOLOGIES & PLATFORMS



Hardware

ESP32, MPU6050, SW-420, HC-SR04, SX1278 LoRa



Embedded

Arduino IDE / C++



Frontend

React, Vite, Tailwind CSS, Recharts



Backend

Node.js, Express.js



Database

Firebase Realtime Database



Alerts

Twilio SMS



Development

Replit

3 MODES OF OPERATION

NORMAL



WARNING



CRITICAL



DEMO



4 IMPACT & FUTURE SCOPE

Impact

- Continuous monitoring
- Faster response to abnormal conditions
- Cost-effective distributed monitoring

Future Scope

- Mine-specific ML datasets and prediction models
- Additional displacement sensors
- 3D ground-movement analysis
- GIS/mine-map integration
- Multiple LoRa gateways
- Mobile and control-room alerts

- **IoT & LoRa-Based Underground Coal Mine Monitoring** – *Microsystem Technologies, 2023*
[Research Paper](#)
- **LoRa Communication for Underground Coal Mines** – *Processes, 2023*
[Research Paper](#)
- **LoRa-Based IIoT Network for Coal Mines** – *IEEE Access, 2024*
[Research Paper](#)
- **IoT/LoRaWAN Wireless Sensor Networks for Mine Monitoring** – *Sensors, 2024*
[Research Paper](#)

Technical References: ESP32 • MPU6050 • SX1278 LoRa • Firebase • React • Node.js/Express • Twilio

Research Foundation:

Sensors → LoRa → Data Analysis → Dashboard → Early Warning